Andrea Capone

PhD Researcher in AI | Software Developer

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Profile

PhD Researcher in Artificial Intelligence with a strong background in software engineering. I work at the intersection of **Machine Learning**, **Formal Reasoning** and **Neurosymbolic AI**, developing methods that reason over incomplete and uncertain information for geophysical risk assessment, in collaboration with **INGV** and **Telecom Paris (France)**. Author of publications at international venues such as **KR** and **EUMAS**, and an active contributor to research software, with hands-on experience building full-stack and data-intensive applications.

Technical Skills

Languages: Python, Java, C

AI & ML: Machine Learning, Time-Series Forecasting, Neurosymbolic AI, Explainable AI, Multi-Agent Systems

Formal Methods: Temporal and Strategic Logics (CTL, ATL), Intuitionistic Logics, Model Checking

Data & Databases: GNSS Time-Series Processing, Signal Processing, Data Analysis, Neo4J, Firebase

Other: Full-Stack Development, Cloud Computing

Publications

- Extended journal version of *An Intuitionistic Version of Alternating-Time Temporal Logic*. Under review, **Theoretical Computer Science**, 2026.
- *Reasoning About Intuitionistic Alternating-Time Temporal Logic*. In Proceedings of the **8th International Workshop on Artificial Intelligence and fOrmal VERification, Logic, Automata, and sYnthesis (OVERLAY)**, 2026.
- *GameL: a Neuro-Symbolic Agentic Architecture Toward Autonomous and Resilient Energy Communities*. In Proceedings of the **41st Italian Conference on Computational Logic (CILC)**, 2026.
- *A Neurosymbolic Approach for Landslide Risk Assessment*. In Proceedings of the **6th National Conference on Artificial Intelligence (Ital-IA)**, 2026.
- *Gaussian Process and Kalman Baselines for Bradyseismic Signal Extraction from GNSS Time Series at Campi Flegrei*. In Proceedings of the **XXXIV Italian Workshop on Neural Networks (WIRN)**, 2026.
- *Machine Learning and Explainable Deep Learning for Forecasting Landslide Deformation using EGMS Time Series*. Preprint, Under Review, **Applied Soft Computing**, 2026.
- *An Intuitionistic Version of Computation Tree Logic*. In Proceedings of the **22nd European Conference on Multi-Agent Systems (EUMAS)**, 2025.
- *An Intuitionistic Version of Alternating-Time Temporal Logic*. In Proceedings of the **22nd International Conference on Principles of Knowledge Representation and Reasoning (KR)**, 2025.

Projects

PhD Project: Neurosymbolic AI for Bradyseism Risk Assessment Nov 2024 – Present
National PhD in AI for Agrifood and Environment *Python, Machine Learning, Formal Methods*

- Extracting the bradyseismic ground-deformation signal from **GNSS time series** at **Campi Flegrei** with **INGV**, comparing **FastICA**, **Gaussian Process** and **Kalman-filter** approaches
- Benchmarking statistical and neural models for **ground-deformation forecasting**, with attention to statistical significance and uncertainty calibration
- Designing **intuitionistic temporal and strategic logics** (ICTL, IATL) with PTIME model checking, to reason about progressively refined and incomplete information

VITAMIN Model Checker Feb 2024 – Present
Logic for Computer Science, Multi-Agent Systems *Python*

- Active contributor to **VITAMIN**, a state-of-the-art model checker for multi-agent systems
- Designed and implemented model-checking procedures for **Intuitionistic Computation Tree Logic (ICTL)** and **Intuitionistic Alternating-time Temporal Logic (IATL)**

• Developing, in collaboration with **Télécom Paris**, a tool that automatically builds **Kripke structures** from **natural-language** scenario descriptions

• Optimized core routines and data-structure management, significantly improving scalability and performance

APLAND 2025 – 2026
AI for Landslide Risk Assessment *Python, FastAPI, Streamlit, Docker*

• Developed a **Frequency Ratio** algorithm for landslide susceptibility, served through a **FastAPI** backend with a **Streamlit** interface and a **Docker** setup simulating a **GeoServer** environment

• Supervised the development of a temporal neural pipeline for **explainable forecasting** of **EGMS** deformation time series, combined with **neural spatial susceptibility mapping**

• Developed a **decoupled neurosymbolic framework** for landslide risk assessment, published at **Ital-IA 2026**

Medical Survey System Feb 2025
Interactive Surveys for Medical Conferences *JavaScript, Firebase*

• Built an **interactive survey platform** for medical conferences, enabling real-time data collection and live visualization

• Integrated **Firebase** for secure authentication, data storage, and real-time synchronization

• Developed responsive **JavaScript** front-end components ensuring smooth usability across devices

Food Talk Jun 2023 – Present
Sustainability, Software Development *Python, Neo4J*

• Designed and developed a **Neo4J**-based system for end-to-end traceability of fruit and vegetable products

• Modeled highly interconnected supply-chain relationships as a graph for efficient querying and analysis

• Exploring partnerships with industry and the University of Naples to scale the platform and advance research

Theater Section Jun 2019 – Feb 2020
Human-Computer Interaction, Software Development *Java, Android, Neo4J*

• Developed a system addressing the exhibition and conservation needs of the **"Fondo Niccolini"** collection

• Used **Neo4J** to model the highly interconnected and complex relationships within the archival data

• Deployed and currently in operation at the **"Certosa di San Martino"**, Naples

Experience

Research and Development Internship Feb 2023 – Nov 2023
Maipek s.r.l. | *University Federico II, Naples, Italy*

• Conducted R&D on an **intraoral scanner**, focusing on 3D reconstruction from optical scan data

• Identified and benchmarked image-processing techniques to build accurate and effective **point clouds**

Education

Ph.D. in Artificial Intelligence for Agrifood and Environment **Status:** Ongoing
National PhD Program in AI | *University Federico II, Naples, Italy* *Nov 2024 – Present*

M.Sc. in Computer Science, curriculum in Reliable Software Systems **Grade:** 110/110 Cum Laude
University Federico II, Naples, Italy *Sep 2020 – Jul 2024*

B.Sc. in Computer Science **Grade:** 94/110
University Federico II, Naples, Italy *Sep 2014 – Mar 2020*

Certifications

• Apple Developer Academy

• Agritech Student Academy

Languages

• Italian – Native Speaker

• English – Professional Working Proficiency

Agreements

I authorize the processing of the personal data contained in my CV pursuant to Article 13 of Legislative Decree 196/2003 and Article 13 of the GDPR (EU Regulation 2016/679).